Topic 4: Materials

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic.

Mathematical skills that could be developed in this topic include determining the slope of a linear graph and calculating or estimating, by graphical methods as appropriate, the area between a curve and the *x*-axis and realising the physical significance of the area that has been determined.

This topic may be studied using applications that relate to materials, for example spare-part surgery.

Students should:
49. be able to use the equation density $\rho = \frac{m}{V}$
50. understand how to use the relationship upthrust = weight of fluid displaced
51. a. be able to use the equation for viscous drag (Stokes' Law), $F = 6\pi\eta rv$. b. understand that this equation applies only to small spherical objects moving at low speeds with *laminar flow* (or in the absence of *turbulent flow*) and that viscosity is temperature dependent
52. **CORE PRACTICAL 4: Use a falling-ball method to determine the viscosity of a liquid.**
53. be able to use the Hooke's law equation, $\Delta F = k\Delta x$, where k is the stiffness of the object
54. understand how to use the relationships • *(tensile or compressive) stress = force/cross-sectional area* • *(tensile or compressive) strain= change in length/original length* • *Young modulus = stress/strain*
55. a. be able to draw and interpret force-extension and force-compression graphs b. understand the terms *limit of proportionality, elastic limit, yield point, elastic deformation* and *plastic deformation* and be able to apply them to these graphs
56. be able to draw and interpret tensile or compressive stress-strain graphs, and understand the term *breaking stress*
57. **CORE PRACTICAL 5: Determine the Young modulus of a material**
58. be able to calculate the elastic strain energy E_{el} in a deformed material sample, using the equation $\Delta E_{el} = \frac{1}{2}F\Delta x$, and from the area under the force-extension graph *The estimation of area and hence energy change for both linear and non-linear force-extension graphs is expected.*